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Supplementary Note 1. Fabrication of PCO_{NA} by solvent-assisted nanotransfer printing combined with pulsed laser deposition

Pr_{0.2}Ce_{0.8}O_{2- δ} (PCO20) was selected as a model MIEC oxide due to its high chemical stability¹, free of elements that tend to segregate to the surface, for example Sr in LSCF and STF²⁻⁴, and the dominance of electronic over ionic conductivity at high pO₂ (e.g. in this study in air)^{5,6}. In conventional solvent-assisted nanotransfer printing (S-nTP), electron beam evaporation is typically used to deposit the target materials, however, this is not suitable for depositing complex oxide materials normally used as MIEC electrodes such as PCO20. Electron beam irradiation, when applied to metal oxides, leads to the reduction and dissociation of the target source into metal and metal suboxide species, each with different volatilities, thereby hindering the reproduction of the stoichiometry of the target source in the deposited film^{7,8}. Pulsed laser deposition (PLD) was therefore selected as an alternative to electron beam evaporation, given its ability to reproduce films with nearly the same stoichiometry as the target source⁹. It is worth noting that this strategy is not limited to PCO used in this study but is extendable to other materials that can be deposited via PLD. We believe that our strategy exhibits wide-ranging applicability and potential for elucidating the precise role of acidity-mediated local space charge across various materials and applications.

Supplementary Note 2. Modeling of heterointerface acidity-mediated space charge effects on electronic properties of PCO_{NA}

Zhang et al. previously showed that decreasing the grain size of a bulk PCO20 sample by lowering the sintering temperature led to a corresponding decrease in conductivity and in E_a from 1 eV down to 0.65 eV. They attributed this to increased strain, that lowers the reduction enthalpies, increases the $[Pr^{3+}]$ concentration, and therefore reduces the conductivity by decreasing the availability of small polaron hopping sites¹⁰. As we show in the Arrhenius plot of **Supplementary Fig. 8**, PCO_{NA} prepared on an Al₂O₃ substrate and annealed at 400 °C instead of 700 °C, exhibited similar E_a and magnitude of conductivity comparable to that of bulk PCO20, despite the sample being composed of nanoscale grains that are presumably smaller due to the lower annealing temperatures.

To rationalize our observations, we note that the activation energy of conductivity (E_a) in bulk PCO is reportedly made up of two terms, given that the electron concentration and mobility are both thermally activated. Following previous derivations¹⁰, $[Pr_{Ce}^x] \sim [Pr_{Ce,tot}]$ is assumed in the high pO₂ regime where overall charge balance is assumed to be $[Pr_{Ce}'] \sim 2[V_{\ddot{O}}]$. Then the small polaron electronic conductivity associated with Pr^{3+} can be written as:

$$\sigma_{Pr^{3+}} = 2^{\frac{1}{3}} [Pr_{Ce,tot}]^{\frac{2}{3}} [O_{\ddot{O}}^x]^{\frac{1}{3}} K_{r,Pr}^0 \mu_{Pr,0}^{\frac{1}{3}} \exp\left(-\frac{\frac{\Delta H_{r,Pr}}{3} + E_{m,Pr}}{k.T}\right) P_{O_2}^{-\frac{1}{6}} \quad (1)$$

where $\Delta H_{r,Pr}$ is the reduction enthalpy of Pr and $E_{m,Pr}$ is the small polaron migration energy. Based on previous literature values, the expected bulk activation energy at high pO₂ and low temperatures is therefore expected to be equal to $\frac{\Delta H_{r,Pr}}{3} + E_{m,Pr} \sim 1$ eV, consistent with our results for PCO_{NA} fabricated on Al₂O₃ substrate annealed at 400 °C (**Supplementary Fig. 8**), where little impact from GB space charge effects was anticipated.

On the other hand, the carrier concentration is also expected to vary spatially according to the magnitude and polarity of the space charge potentials at both heterointerfaces and GBs. Since small polarons are mobile, they can readily redistribute within the space charge region in response to changes in space charge potential, similarly to the oxygen vacancies, a situation that can be described by the Gouy-Chapman approximation¹¹, where a flat spatial profile of $[Pr_{tot}]$ is assumed. To a first-order approximation, the variation in Pr_{Ce}' adjacent to the interface in response to the space charge potential profile can be defined as¹²: